Diagnostic Report generated by gemini-2.5-pro

Diagnostic Analysis of Run_23

1. Summary of Key Deviations

The analysis of Run_23 reveals a critical and highly isolated sensor failure with no significant secondary physical effects.

- **p_FMU (Critical):** This is the primary and most severe anomaly. The pressure sensor at the Fuel Metering Unit shows a **major negative drift** beginning in state s2, which worsens to a **complete signal dropout (-99.4%)** in state s5. The signal partially recovers but remains at a major negative offset for the rest of the run. This is a classic sensor failure signature.
- **Flow_Bypass & Flow_Pump (Minor):** Both of these flow signals show a slight **positive offset** (around +9% and +4% respectively). These are minor and likely represent either a slight system over-performance or minor calibration drifts.
- **All Other Signals (Nominal):** Most critically, all `Flow_Engine` signals are completely healthy, showing no significant deviations. All other pressure signals (`p_Pump`, `p_Combustion`, `p_Shut`) and `Flow_PRV` are also nominal.

2. Root-Cause Hypotheses

Rank	Suspected Component	Fault Type	Supporting Evidence
1	**p_FMU Sensor**	**Sensor Dropout / Total Failure**	This is the most probable cause. The `p_FMU` sensor signal drops to nearly -100%, a clear sign of failure. The most compelling evidence is the complete lack of physical consequences: a real pressure loss at the FMU would starve the engines of fuel, causing all `Flow_Engine` signals to plummet. Instead, the engine flows remain perfectly healthy, proving the physical system is fine and only the sensor has failed.

2	**Wiring / Electrical Fault to p_FMU**	**Signal Loss / Grounding Issue**	This is physically equivalent to a sensor failure. A break in the wiring or a grounding fault in the `p_FMU` sensor's circuit would result in a lost signal, explaining the dropout reading without any actual change in system pressure.
3	**Fuel Metering Unit (FMU)**	**Major Internal Leak**	A massive leak *within* the FMU could theoretically cause a localized pressure drop at that specific sensor. However, such a large leak would divert flow and almost certainly affect the downstream `Flow_Engine` readings, which are perfectly healthy. Therefore, a physical leak is highly unlikely.
4	**Main Fuel Pump**	**Minor Over-performance**	The pump could be delivering ~4% more flow, explaining the minor positive offset on `Flow_Pump`. This is a possible, but very minor, secondary issue that is completely overshadowed by and unrelated to the critical `p_FMU` failure.
5	**System Contamination**	**Debris affecting p_FMU sensor port**	It is conceivable that a piece of debris could have blocked the sensing port for the `p_FMU` sensor, causing it to read incorrectly. This is a possible underlying cause for the sensor failure itself.

3. Most Probable Hypothesis

The most probable root cause is a **Total Sensor Failure of the p_FMU Sensor**, likely due to an internal fault or an electrical/wiring issue.

Reasoning: The evidence for an isolated sensor failure is definitive. The `p_FMU` reading drops to nearly zero, but this has **no effect on the downstream engine flows**. The `Flow_Engine1-4` signals all remain nominal, which would be physically impossible if the pressure at the FMU had actually vanished. The system continues to operate as if nothing is wrong, proving that the pressure is still there and only the measurement is lost. The other minor flow deviations are likely unrelated sensor drift or noise and are not part of the primary fault.